Table 64. USART (SPI mode) characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CK}	USART clock frequency	Master mode $2.0\ V < V_{DD} < 3.6\ V$ $1.65\ V < V_{DDIO2} < 3.6\ V$	-	-	6.0	MHz
		Slave receiver mode $2.0\ V < V_{DD} < 3.6\ V$ $1.65\ V < V_{DDIO2} < 3.6\ V$	-	-	16.0	
		Slave transmitter mode $2.0\ V < V_{DD} < 3.6\ V$ $1.65\ V < V_{DDIO2} < 3.6\ V$	-	-	16.0	
$t_{su(NSS)}$	NSS setup time	Slave mode	T_{ker}(1) + 1	-	-	ns
$t_{h(NSS)}$	NSS hold time	Slave mode	2	-	-	ns
$t_{w(CKH)}$	CK high time	Master mode	$1 / f_{CK} / 2 - 1.5$	$1 / f_{CK} / 2$	$1 / f_{CK} / 2 + 1.5$	ns
$t_{w(CKL)}$	CK low time					
$t_{su(RX)}$	Data input setup time	Master mode $2.0\ V < V_{DD} < 3.6\ V$ $1.65\ V < V_{DDIO2} < 3.6\ V$	19	-	-	ns
		Master mode $2.0\ V < V_{DD} < 3.6\ V$ $2.0\ V < V_{DDIO2} < 3.6\ V$	16	-	-	
		Slave mode	1.5	-	-	
$t_{h(RX)}$	Data input hold time	Master mode	0	-	-	ns
		Slave mode	0	-	-	
$t_{v(TX)}$	Data output valid time	Slave mode $2.0\ V < V_{DD} < 3.6\ V$ $1.65\ V < V_{DDIO2} < 3.6\ V$	-	13.5	25.5	ns
		Slave mode $2.0\ V < V_{DD} < 3.6\ V$ $2.0\ V < V_{DDIO2} < 3.6\ V$			22	
		Slave mode $2.7\ V < V_{DD} < 3.6\ V$ $2.7\ V < V_{DDIO2} < 3.6\ V$	-		17.5	
		Master mode	-	2.0	4	
$t_{h(TX)}$	Data output hold time	Slave mode	11.5	-	-	ns
		Master mode	0.5	-	-	

1. T_{ker} is the *usart_ker_ck_pres* clock period